

KRASNODAR, Russian FederationWMO: **349290**Lat: **45.033N**Lon: **39.150E**Elev: **34**StdP: **100.92**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.1	-10.3	-17.1	0.8	-13.8	-13.4	1.2	-9.8	11.9	4.1	10.3	0.4	4.2	70	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	34.9	22.5	33.1	22.0	31.3	21.4	24.2	31.2	23.3	30.1	22.5	29.0	4.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	16.8	27.9	21.1	15.8	26.7	20.2	14.9	25.6	73.4	31.0	69.6	30.4	66.4	28.8	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.4	9.2	8.3	-18.6	37.4	6.5	1.9	-23.2	38.8	-27.0	39.9	-30.6	41.0	-35.3	42.5	
(5)				-18.9	25.8	6.4	1.0	-23.5	26.6	-27.2	27.2	-30.8	27.8	-35.4	28.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	0.3	1.6	6.2	12.0	17.6	21.8	24.5	24.4	19.1	12.4	6.4	2.1
	DBStd	9.60	5.63	5.52	4.11	3.76	3.60	2.95	2.66	3.07	3.68	4.44	4.42	5.13
	HDD10.0	1086	302	237	128	21	1	0	0	0	1	26	124	247
	HDD18.3	2743	559	468	376	191	59	6	0	1	34	187	359	504
	CDD10.0	1975	1	2	12	82	236	355	449	447	273	102	15	2
	CDD18.3	590	0	0	0	2	35	111	192	189	57	5	0	0
	CDH23.3	5841	0	0	1	34	333	981	1971	1972	493	55	1	0
	CDH26.7	2343	0	0	0	4	80	341	861	899	149	9	0	0
Wind	WSAvg	3.7	3.9	4.2	4.5	4.2	3.6	3.4	3.4	3.5	3.4	3.4	3.5	3.6
Precipitation	PrecAvg	702	62	42	45	58	68	81	56	53	44	46	66	77
	PrecMax	1024	158	138	118	192	158	307	154	178	108	91	122	167
	PrecMin	438	7	1	0	10	21	5	2	1	1	1	4	8
	PrecStd	124	37	32	25	37	34	56	42	43	28	23	36	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	17.0	21.0	26.9	30.9	34.8	36.9	37.2	33.8	27.9	21.2	15.3
		MCWB	8.8	10.3	12.1	16.1	20.4	21.8	22.7	22.0	21.8	18.7	13.7	10.4
	2%	DB	10.8	13.8	17.8	23.9	28.9	32.1	34.9	35.1	30.2	24.8	17.8	12.2
		MCWB	7.4	9.0	11.1	15.1	19.5	21.6	23.0	22.3	20.0	17.2	12.6	8.9
	5%	DB	8.8	10.9	15.0	21.4	27.0	30.1	33.0	33.2	28.2	22.0	15.2	10.6
		MCWB	6.4	7.6	9.8	14.1	18.6	21.1	22.4	21.9	19.3	16.0	11.7	8.0
	10%	DB	6.9	8.8	12.8	19.1	25.0	28.2	31.1	31.7	26.1	19.5	12.9	8.9
		MCWB	5.3	6.2	8.7	12.9	17.8	20.2	21.6	21.4	18.4	14.9	10.2	7.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	11.1	13.0	17.5	21.8	24.2	25.7	25.5	22.7	19.5	14.7	11.1
		MCDB	12.5	15.4	18.9	24.5	28.7	30.8	33.2	32.9	30.8	25.6	18.9	14.3
	2%	WB	7.8	9.4	11.6	15.9	20.4	23.1	24.5	24.1	21.3	17.9	13.3	9.5
		MCDB	9.9	13.2	16.8	22.1	27.0	29.6	31.5	31.4	27.8	23.1	16.7	11.9
	5%	WB	6.3	7.7	10.2	14.7	19.4	22.2	23.6	23.1	20.4	16.6	11.8	8.2
		MCDB	8.1	10.3	14.3	20.3	25.5	28.1	30.2	30.1	26.3	21.0	15.0	10.3
	10%	WB	5.1	6.2	8.9	13.5	18.3	21.2	22.6	22.3	19.3	15.4	10.4	6.9
		MCDB	6.6	8.3	12.2	18.3	23.8	26.6	29.0	28.9	24.6	19.0	12.7	8.5
Mean Daily Temperature Range	5% DB	MDBR	6.7	8.2	8.9	10.8	11.0	11.1	11.7	12.3	11.5	10.1	8.4	6.7
		MCDBR	9.1	11.8	13.6	14.7	14.2	13.6	14.1	14.4	14.4	14.3	12.8	9.8
		MCWBR	6.5	7.8	8.1	7.7	6.8	5.9	5.7	5.9	6.6	7.8	8.2	6.9
	5% WB	MCDBR	8.4	10.8	12.6	12.9	12.9	11.7	12.3	12.6	12.6	12.9	11.7	9.0
		MCWBR	6.3	7.5	7.8	7.1	6.5	5.5	5.7	5.7	6.3	7.5	7.9	6.6
Clear-Sky Solar Irradiance	taub	0.348	0.379	0.422	0.459	0.425	0.430	0.454	0.452	0.417	0.390	0.358	0.347	
	taud	2.279	2.205	2.119	2.061	2.251	2.274	2.220	2.221	2.292	2.327	2.352	2.301	
	Ebn at Noon	738	782	798	802	847	842	816	801	798	765	725	695	
	Edh at Noon	84	109	136	156	134	131	137	132	113	96	78	75	
All-Sky Solar Radiation	RadAvg	1.24	2.11	3.04	4.67	5.85	6.40	6.47	5.79	4.27	2.72	1.56	1.05	
	RadStd	0.19	0.27	0.28	0.49	0.61	0.40	0.34	0.39	0.41	0.29	0.20	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (7 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+134	+70	